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E-210 LOW VOLTAGE MOTORS

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to the site, installation, commissioning and testing of all low voltage motors (rated at 400 V, 3 phase or 230 V, single phase, 50Hz, or 220 V DC) in accordance with Part 4, Subsection 1.2 Summary of Work.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards and the following Specific Standards.

2.1 National

PUIL : Peraturan Umum Instalasi Listrik - Indonesia (General Regulation of Electrical Installation Indonesia).

SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

IEC 60034 : Rotating electrical machines

IEC 60072 : Dimensions and output series for ratings for electrical machines

IEC 60079 : Electrical apparatus for explosive gas atmospheres.

IEC 60445 : Basic and safety principles for man-machine interface, marking and identification of equipment terminals and of terminations of certain designated conductors, including general rules for alphanumeric system.

2.3 American National Standards Institute and Institute of Electrical and Electronics Engineers

ANSI/IEEE - 112 : Test Procedures for Polyphase Induction Machines.

ANSI/IEEE - 113 : Test Procedures for Direct Current Machines.

ANSI/IEEE – 114 : Test Procedures for Single Phase Induction Motors.

2.4 National Electrical Manufacturers Association

NEMA MG 1 : Motors and Generators

2.5 National Fire Protection Association

NFPA 70 : National Electrical Code

2.6 Anti Friction Bearing Manufacturer's Association

AFBMA : Anti Friction Bearing Manufacturer's Association.

3. OPERATING AND DESIGN CONDITIONS

Motors shall be suitable for operation in a combined cycle power plant located in a tropical climate along the seacoast. Refer to Part 4, Subsections 1.1, 2.14 and 2.16 for the meteorological, seismic and environmental conditions at the Site.

Motors shall deliver their rated kW output, without damage, with $\pm 10\%$ variation of rated voltage at the terminals at rated frequency. Motors shall deliver their rated kW output, without damage, with $\pm 5\%$ variation in frequency.

Motors shall deliver their rated kW output, without damage, when the sum of the simultaneous variation of voltage and frequency does not exceed $\pm 10\%$.

4. DETAILED REQUIREMENTS

4.1 General

Motors shall be heavy duty, having highest possible efficiency and power factor compatible with good motor design.

Motors 10 kW and larger shall be supplied with anti-condensation heaters, suitable for 230 V AC, single phase 50 Hz supply. Heaters shall be automatically de-energized when the motor is switched on and automatically energized when the motor is not operating.

The front end of the shaft (opposite the driving end) shall be centered and shall be accessible for taking speed readings with a portable tachometer.

Motors shall be designed, built and adequately braced for across-the-line starting with 80 to 110% of rated voltage applied at the motor terminals.

4.2 Ratings

All AC motors up to and including 750W may be rated either 230V, single phase, 50 Hz, or 400V, 3 phase, 50 Hz, and shall be supplied from power panel or motor control centers (MCCs).

All AC motors up to and including 60 kW shall be 400 V, 3 phase, 50Hz and shall be supplied from MCC. All AC motors from 60 kW up to and including 150 kW shall be 400V, 3 phase, 50 Hz, and shall be supplied from PDC switchgear.

All AC motors larger than 150 kW shall be medium voltage motors.

All DC motors which will operate from the station batteries shall be rated 220 V DC. All DC operated valve motors shall be rated for operation from + 15% to -30% of rated voltage. All other DC motors shall be suitable for continuous full load operation from + 15% to -15% of rated voltage.

The motors shall have 1.0 service factor. All motors shall be rated for not less than 115% of the requirements of the driven equipment when operating under design requirements and not less than 100% of the power required to operate the driven equipment at its maximum capability.

Motors shall be rated for Continuous operation in a 40°C ambient temperature with a motor winding temperature rise by resistance not exceeding 80°C.

All motors shall be of a standard size in accordance with IEC 60072.

4.3 Insulation

Motor insulation shall be Class F with Class B insulation temperature rise.

4.4 Torque Requirements

Motors shall be designed to have sufficient torque for starting and accelerating the maximum load when 80% of motor rated voltage is applied to the motor terminals, without exceeding the thermal limits of the motor. The permissive locked rotor time shall be greater than the accelerating time, with the motor loaded and with motor terminal voltage at 80% rated voltage.

Breakdown torque shall be adequate to maintain motor operation during a voltage dip to 75% of rated voltage for period of 60 seconds at rated load and frequency.

Squirrel-cage induction motors, operating with rated terminal voltage and rated frequency, shall have torque values in accordance with the applicable IEC standards and/or requirements determined by the nature of driven equipment with minimum starting voltage and voltage dips, while the motor is running.

Motors shall have the lowest locked rotor current values consistent with high efficiency and good performance. However locked rotor current, in percent of full load current, shall not exceed 650%.

4.5 Enclosures

Motors located indoors shall be totally enclosed fan cooled or totally enclosed non-ventilated with a minimum degree of protection of IP 55 as per IEC 60034.

Motor required for driving submersible pumps shall be suitable for operation under conditions specified in IEC 60034 for enclosure type IP 58.

Motor located outdoors shall be totally enclosed fan cooled. Outdoor motors shall be provided with seals at the shafts, terminal box, and around motor terminal leads, to prevent ingress of water.

Motors located in hazardous areas as defined in NFPA 70 shall be explosion proof or dust-ignition proof as required.

Housings shall be cast iron or steel plate. Aluminum shall not be used.

4.6 Bearings

Ball or roller anti-friction bearings shall be provided in accordance with manufacturer's best design for the motor application, and shall be sealed in dust tight housings to exclude foreign matter.

Anti-friction bearings for direct coupled motors shall have minimum L-10 life of 150,000 hours in accordance with AFBMA.

Motors with anti-friction bearings shall be provided with vibration detectors in accordance with Part 4 Technical Specification I-001, where required. Refer to Clause 4.7 herein for termination requirements.

4.7 Terminal Boxes

Terminal box for motor leads shall be waterproof, corrosion resistant. Motors, 100 kW and larger, shall be provided with non-magnetic gland plates.

All terminal boxes shall be positively grounded to the motor housing and also they shall be provided with grounding terminals inside the terminal box housings for instrument cable shield grounding.

4.8 Winding Temperature Detectors

Motors, 100 kW and larger, shall be provided with stator winding temperature detectors, which shall consist of six (6) resistance temperature detectors (RTDs), two (2) RTDs per phase, embedded in the stator windings.

4.9 Direction of Rotation and Terminal Markings

The connections to AC motors shall be such that when the supply leads L1, L2 and L3 are connected to motor terminals, U,V,W respectively, the motor shall rotate in a clockwise direction, when viewed from the driving end. Where such motors require anti-clockwise rotation, the supply leads L1, L2 and L3 shall be connected to terminals U, W and V respectively, in accordance with IEC 60445.

4.10 Nameplate

The nameplate shall list all motor data required by IEC 60034 and the followings:

- (1) Service factor
- (2) Shaft rotation corresponding to a supply voltage sequence of U-V-W
- (3) Lubricant type and lubrication instructions
- (4) Terminal arrangements
- (5) Maximum permissible number of starts and the required cooling period when the motor is started under cold and hot motor conditions for motors 100 kW and larger.

4.11 Service and Life

The motors shall be designed for a thirty-year service life and operating frequency compatible with that of the driven equipment.

4.12 Motor Operated Valves and Dampers

4.12.1 General

The motor-operated valve control shall include the motor, operator unit gearing, position switch gearing, local valve position indicator, provisions for manual operation, and torque and position switches. When the stem driving mechanism of the standard valve is not utilized, the operator shall include a stem nut for rising stem valves or a bored and keyed drive sleeve for not-rising stem valves. Stem nuts shall have a positive locking feature to prevent loosening during valve travel. Starters for motor operated valves shall be remotely located in their associated motor control center.

The control shall be capable of fully seating the valve (leaktight) under electrical operation without jamming or requiring extremely accurate position switch adjustment. Gate and globe valves shall be seated under torque control. Butterfly, plug, and parallel disc gate valves shall be seated under position switch control.

A means of preventing overtravel due to inertia of the drive shall be provided and the means shall be operative in stopping the valve at the open, closed or any intermediate position.

Manual operation of the valve shall be by a handwheel. The operator shall stay in the manual position until energized electrically. The motor shall not rotate during hand operation. A failed motor or motor bearing shall not prevent manual operation.

Valve stem covers shall be provided.

Ball or roller bearings shall be used throughout the main power train.

All gearing shall have a self-locking design, to prevent relaxation of the valve gearing after torque seating, except where unusually fast valve operation is required, which would prohibit this feature. When furnished, nonlocking gears shall be positively indicated on the valve operator drawing.

4.12.2 Valve Motors

The motor shall be high torque, totally enclosed, non-ventilated, specifically designed for valve operator service. All AC Motor operators shall be rated 400V, 3 phase, 50 Hz.

The motor shall be pre-lubricated. All bearings shall be of the anti-friction type. Bearings shall be so mounted as to facilitate replacement and maintenance. If lubrication is required during the life of the motor, it shall be accomplished without removal of bearing caps. No special lubrications shall be required up to an ambient temperature of 65°C.

The motor shall be sized to open or close the valve against the maximum specified differential pressure, without damage, when the voltage at its terminals varies $\pm 10\%$ of rated voltage for AC and $+15\%$ to -30% of rated voltage for DC motors.

Insulation shall be Class F with a Class B temperature rise, unless otherwise specified or anticipated environmental conditions require a different class.

4.12.3 Stem Nut

The valve or valve control shall have a stem nut of high tensile bronze, ductile iron, or other material compatible with the valve stem.

4.12.4 Limit and Torque Switches

Sixteen geared limit switches shall be furnished. The limit switches shall serve to stop the valve travel where appropriate at fully open or fully closed, to bypass torque switches, to operate indicating lamps that shall show valve position and to operate sequencing or interlock circuits where appropriate.

Limit switches shall be mechanically connected to the moving valve stem and shall be in step, at all times, while the valve is being operated electrically or mechanically. Operation of the handwheel shall not change the set points of the switches.

Torque switches shall be furnished for stopping the motor in its extreme operating positions and to protect the motor if foreign objects are encountered during either a closing or opening operation. They shall be separately adjustable.

4.12.5 Position Indicator

A mechanically operated valve position indicator shall be provided on each operator to give a positive indication of position at all times.

4.12.6 Brake

If required for satisfactory operation of the valve, a spring applied, solenoid release brake shall be provided. The drum shall be directly coupled to the motor shaft.

4.12.7 Position Transmitter

A position transmitter shall be furnished for valves or dampers of the modulating type. Equipment shall be of the electrical type, suitable for operation at 110 volts, 50 hertz, single-phase, AC. The transmitter shall be totally enclosed, and shall be directly connected or geared to the valve stem, and it shall remain in synchronism during both hand and motor operation. The output signal shall be 4-20 mA DC.

4.12.8 Wiring

Control and power wiring shall be a minimum of 2.5 mm² except where larger size conductors are needed for current carrying requirements. The conductors shall be stranded copper.

5. PERFORMANCE TESTING AND GUARANTEES

Standard type tests shall be performed on one (1) motor of each type and rating for AC and DC motors, in accordance with test procedures described in IEC Standard. The Contractor may submit test certificates of type tests which have been carried out on identical equipments

Routine factory tests shall be performed on all the rest of the motors, in accordance with the applicable parts of the above IEC Standards.

Certified test reports, shall be submitted to the Owner for review.

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6. QUALITY ASSURANCE/QUALITY CONTROL

The Contractor shall fulfill the quality assurance/quality control requirements listed for individual equipment in the respective equipment Specifications and in Part 4, Section 2.7.

The quality assurance/quality control procedures shall be as specified. These shall include, but not be limited to, the followings:

- (1) Insulation and High Voltage Tests.
- (2) Cleaning and Painting.

The quality assurance/quality control documentation as specified shall include, but not be limited to, the followings:

- (1) Inspection Reports
- (2) Certified Test Reports.
- (3) All other documentation required by applicable codes and standards.

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